Technology Stack

Date: 29 june 2026	
Team ID: *(xxxxxx)*	
Project Name: Human Development Index (HDI) Prediction Using Machine Learning	

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	**HTML5, CSS3, Bootstrap, JavaScript**	Used to build a responsive and userfriendly interface where users can enter HDI indicators and view prediction results.
2	Backend / Server-Side	**Python 3.x, Flask Framework**	Python provides machine learning support, while Flask connects the trained model with the web interface and processes user requests.
3	Database / Data Storage	**CSV Dataset, Pandas DataFrame, Joblib (.pkl)**	The HDI dataset is stored in CSV format for training. The trained model is saved as a .pkl file using Joblib for quick prediction.
4	Cloud / Hosting / Deployment	**Flask Local Server (Development)** *(Future: Render / Railway / PythonAnywhere)*	The application runs on a local Flask server during development and can be deployed to a cloud platform in the future.
5	Version Control & CI/CD	**Git, GitHub**	Git tracks project changes, while GitHub enables version control, collaboration, and project backup for all team members.

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6	Third-Party APIs / Other Tools	**Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Anaconda, VS Code**	Scikit-learn is used to build the machine learning model, Pandas and NumPy handle data processing, Matplotlib and Seaborn generate visualizations, Jupyter Notebook is used for model development, Anaconda manages the Python environment, and VS Code is used for coding and debugging.

Team Members and Responsibilities

Team Member	Role	Responsibility
G Naveen	Team Lead	Project planning, GitHub repository management, model integration, and coordination of team activities.
Chandana Busireddy	Member	Dataset collection, preprocessing, and preparation of project documentation.
Gandham Joshna	Member	Exploratory Data Analysis (EDA), charts, graphs, and data visualization.
Nandineni Gowthami	Member	Flask web application development, HTML interface design, and application testing.
Shaik Talha	Member	Machine learning model training, model evaluation, GitHub updates, report preparation, and final project presentation.

Why This Technology Stack?

- **Python** provides powerful libraries for machine learning and data analysis.
- **Flask** creates a lightweight and efficient web application.

- **Scikit-learn** offers reliable algorithms for HDI prediction.
- **Pandas** and **NumPy** simplify data preprocessing and numerical computations.
- **Matplotlib** and **Seaborn** help visualize trends and relationships in HDI data.
- **HTML, CSS, and Bootstrap** create an intuitive interface for users.
- **GitHub** ensures secure code management and teamwork.

Expected Outcome :-

The selected technology stack enables the team to build a reliable, scalable, and easy-to-use **Human Development Index Prediction System**. The application allows users to enter development indicators, processes the data through a trained machine learning model, and predicts whether a country belongs to the **Very High**, **High**, **Medium**, or **Low** HDI category. This solution supports governments, researchers, students, and policymakers in making informed, data-driven decisions.